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Unstructured data analysis report

-Most of the data is unstructured, like videos and written documents.

-AI vs. lawyers: they had to work with large amounts of unstructured data, and AI achieved higher accuracy than the lawyers.

-The video then talks about text mining. It consists of extracting interesting and non-trivial information and knowledge from unstructured text; it's the process used to filter spam emails by finding keywords often used in scams such as jackpot or loan, and it can also extract the main topics in large amounts of data.

-Scientific publications show that older methods based on logic or strict rules are declining. Today, the field is dominated by Machine Learning, Neural Networks, and Reinforcement Learning.

-Text analytics has many applications, such as summarization, for example creating word clouds or finding keywords in different searches. It's also used for Categorization: predicting if a movie review is good or bad using many keywords. Recommendation: improving suggestions and filtering out fake reviews.

-It's also used in finding restaurants by analyzing reviews and comments on the restaurant's posts.

-Unstructured data can be used with structured data to improve prediction model performance. It helps predict movie box office scores based on social media posts.

-Text analysis can encounter many problems, like the complexity of human language. Words can have many meanings (like Apple can be the company or the fruit), which is why context is key to understanding. The other challenge is the fact that text analysis involves managing huge amounts of documents and information.

Unstructured data can take 2 forms: weakly structured (research papers, news stories, etc.) and semi-structured (emails, HTML, PDF files, etc.).

A simple process of Text Analytics is to first mine and collect data from the internet or a digital library. Step 2 is to transform the data into a structured format, such as converting text into a table with word counts. Step 3 is to select and extract features, like identifying key terms. The final step is to use algorithms and evaluation to track the performance of the model, to check if it reaches the initial objectives.